How Moirai-1.1 makes a forecast

Salesforce · 14M parameters (small) · masked encoder with any-variate attention · zero-shot

1. All rows in one sequence

Targets and covariates are laid end to end as one long sequence. A ? is a value not known yet.

past *future*

Target temperature

Known ahead day length

Past only NAO index

target **known** **past**

joined into one sequence

Any number of rows fits, so the five metals share one sequence.

2. Scale, patch, mask the future

Each row is scaled by the mean μ and standard deviation σ of its own past.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

value

one patch = 32 steps

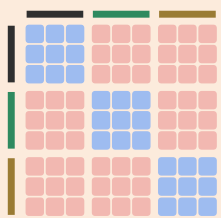
residual MLP

one token: 384 numbers

Patch sizes 8 to 128 each have their own input layer; our runs use 32. Future target patches become a learned mask token M.

3. Six blocks of any-variate attention

Every token reads every other token, past and future, in every row. RMSNorm in front, shortcut around.



same row other row

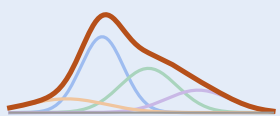
Time: rotary positions (RoPE) along each row

Rows: one learned bias for same row, one for other row; no row order

Feed forward: SwiGLU, 384 → 1,024 → 384

4. A mixture, then samples

Each mask token gives, for each of its 32 steps, a **mix of four distributions**: Student-t, normal, negative binomial, log-normal.



one step: four curves mixed

median

sample paths

100 paths are drawn; their median is the point forecast.

Any-variate attention takes any number of rows, so covariates and related series go in as they are.

6 blocks · width 384 · 6 heads · patch sizes 8 to 128 · the whole horizon in one pass

In the notebooks: last 2,048 days or full monthly history · all covariates · five metals together · patch 32 · 100 samples · CC-BY-NC-4.0